

Machine Learning

Course syllabus. ML Paradigms. ML History

Aleksandr Petiushko

ML Research



Content

① Course syllabus

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- ② ML Paradigms

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- ① Course syllabus
- ② ML Paradigms
- ③ Historic reference

What is Artificial Intelligence?

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- Able to perceive the information, analyze it, make decisions based on this analysis

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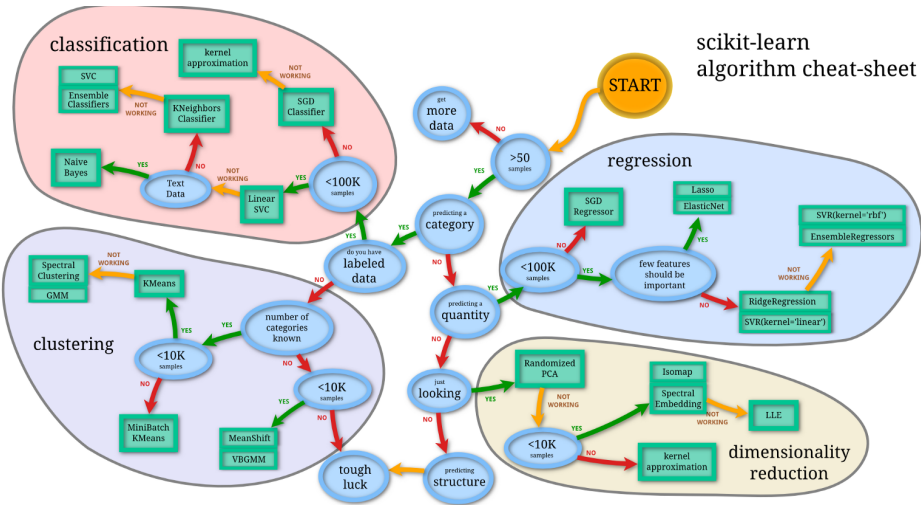
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Artificial Intelligence

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- (**General**) The same as natural intelligence, but computer is instead of human
- (**Weak**) Algorithm which is able to be trained using the input data in order to do tasks afterward — instead of human

Scikit-Learn¹ Roadmap



¹https://scikit-learn.org/stable/tutorial/machine_learning_map/

(Tentative) future content

Theoretic part

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Practice part

- Data processing and analysis by Python
 - Scikit-Learn, Numpy, ...

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General definition

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In 1997 Tom M. Mitchell introduced more formal definition of a machine learning algorithm.

Formal definition

A **computer program** is said **to learn** from examples E for some set of problems T and a quality metric P if its performance on problems from T , as measured by P , is improved by using examples E .

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- **1901**: Karl Pearson invented the Principal Component Analysis (PCA) — a master method for data dimensionality reduction.
- **1906**: Andrey Andreyevich Markov develops the apparatus of Markov chains, which in **1913** he uses to study the text “Eugene Onegin”. Markov chains are used to generate and recognize signals.

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- **2022:** OpenAI, a (not so) non-profit research company, provided the breakthrough in LLMs: ChatGPT.

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- Reinforcement
 - Action generation based on interaction with the environment

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Thank you!